

Dataset Links and Usage Document

CSE4078 Spring 2026 Term Project – Group 6

Turkish Legal Question Answering with Small Language Models

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Dataset inventory summary. The project used one external dataset. The official training and test splits of this dataset were preserved, while the training split was further divided into project-specific training and validation subsets. No additional external corpus was used for model training or evaluation.

Dataset Used

Required Item	Dataset Information
Dataset name	Turkish Law Chatbot Dataset
Dataset identifier	Renicames/turkish-law-chatbot
Source	Hugging Face Datasets Hub; published under the account Renicames.
Dataset link	https://huggingface.co/datasets/Renicames/turkish-law-chatbot
Short description	A Turkish text question-answering dataset focused on legal topics, including constitutional law, legal concepts, procedures, rights, obligations, and scenario-based legal questions. Records contain a Turkish question and a corresponding reference answer.
Task / modality	Question answering; text; Turkish; distributed in JSON-compatible form through the Hugging Face dataset interface.
License	Apache License 2.0, as stated on the Hugging Face dataset page.
Official split sizes	13,354 training examples and 1,500 test examples, for a total of 14,854 examples.
How it was used	The official training split supplied the supervised fine-tuning corpus and the validation subset. The official test split was used as the common, held-out evaluation corpus for both baseline models and the final fine-tuned model.

Project-Specific Use of the Dataset

Supervised Fine-Tuning Corpus

The official training split was loaded with the Hugging Face `datasets` library. Each example was formatted using a Turkish legal-assistant prompt containing a system instruction, question, optional context, and reference answer. Empty formatted records were removed and exact duplicate formatted examples were checked before splitting. No exact duplicates were removed in the final run, so all 13,354 official training records remained available.

Training and Validation Subsets

The retained official training split was divided with a fixed random seed of 42:

Derived subset	Examples	Share	Project use
SFT training subset	12,018	90%	QLoRA supervised fine-tuning
Validation subset	1,336	10%	Validation loss and checkpoint selection
Official test split	1,500	Held out	Common baseline and final evaluation

Common Test Corpus

The official 1,500-example test split was not used during training, preprocessing decisions, validation, or checkpoint selection. The same ordered test corpus was used to evaluate:

- the baseline `Qwen/Qwen2.5-3B-Instruct` model,
- the baseline `amd/Instella-3B-Instruct` model, and
- the QLoRA-fine-tuned Qwen model.

All reported comparisons used the same questions and reference answers. Generated answers were stored as JSON Lines records containing the model name, question, reference answer, and generated answer. ROUGE and BERTScore were then computed from these saved records.

Data Handling and Reproducibility Notes

- The dataset was downloaded programmatically with `load_dataset("Renicames/turkish-law-chatbot")`.
- The project did not merge the corpus with any additional legal dataset.
- The official test split remained isolated until baseline and post-training evaluation.
- The same test split was used for every reported model comparison.
- Project-generated inference files and metric files are experimental outputs, not additional datasets.
- The dataset includes some noisy or mismatched question-answer pairs; this limitation was considered in the qualitative error analysis.

Repository and Access Links

Resource	Link
Dataset page	Hugging Face: Renicames/turkish-law-chatbot
Project repository	GitHub: cereneren/CSE4078S26_Grp6